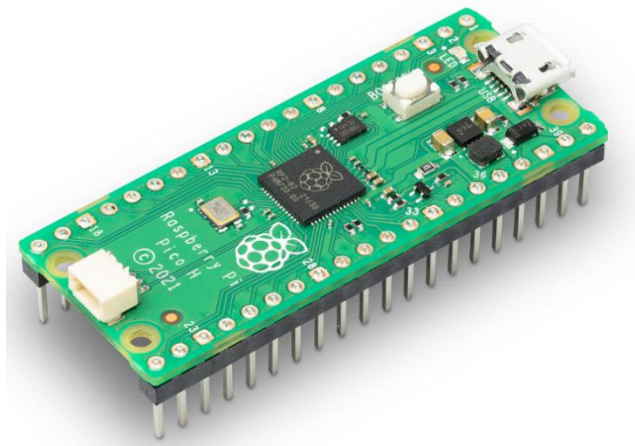


RISER Embedded System LAB Experiments



Visteon®

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1. Print a message in UART console.

Write a simple C program to print a message in a UART Serial Console.

- i. Prerequisites
 - a. Pico microcontroller
 - b. Putty Serial console (In PC).
- ii. Connection Details
 - Connect the Power cable to board and check USB serial device enumerated in Device manager of your PC. Now open the putty application set the COM port and baud rate as 115200. Please note that your power cable also has USB to Serial connection.
- iii. Expected Output
 - Print “Hello Riser” in the Uart console.

2. Read and write the register values of a Microcontroller.

Write a simple C program to have menu driven option to read and write a registers of the microcontroller.

- i. Prerequisites
 - a. Pico microcontroller
 - b. Putty Serial console (In PC).
- ii. Connection details
 - Check COM port and open in putty serial console.
- iii. Expected Output
 - Enter ‘1’ to read register value.
 - Ask the Register address from user and print the value of register.
 - Enter ‘2’ to write register value.
 - Ask the register address and value from user. Write the given value to given register.

3. Adjusting LED brightness by using Potentiometer

Write an application program in C to calculate voltage level by using potentiometer and with that voltage value adjust the LED brightness.

- i. Prerequisites
 - a. Pico microcontroller
 - b. Bread board
 - c. LED
 - d. Potentiometer
- ii. Connection Details
 - Potentiometer connected ADC pins
 - LED connected with PWM pins of Mico
- iii. Expected Output
 - Based on Potentiometer adjustment LED brightness should varies. For brightness adjustment whatever values read from the Potentiometer which loaded to PWM pin(register) to adjust the LED brightness.

4. Identify key press in keypad matrix and print the scan value to the console

Write a Keypad scanning program and display the Key press value to console whenever user press the any of the key in Matrix keypad.

- i. Prerequisites
 - a. Pico microcontroller
 - b. Keypad Matrix
 - c. Putty Serial console (In PC).
- ii. Connection details
 - Keypad matrix connected GPIO pins.
 - Check COM port and open in putty serial console.
- iii. Expected Output
 - On every Key press on keypad matrix, the value should print in UART serial console.

5. Ultrasonic sensor

Write a small application program to print the distance value in the serial console from the nearest object to your sensor.

- i. Prerequisites
 - a. Pico microcontroller
 - b. Ultrasonic Sensor
 - c. Bread board
- ii. Connection details
 - Ultrasonic sensor Trig and Echo pins connected with GPIO Pins
- iii. Expected Output
 - Continuously print the distance values of the nearest object which around the sensor. Demonstrate the distance value increase or decrease when the object moves.

6. NOR Memory Access

Write a C program to erase, write and read the NOR memory and print the read result.

- i. Prerequisites
 - a. Pico Microcontroller
 - b. W25Q64 (NOR device)
 - c. Bread board
- ii. Connection details
 - Connect the NOR device with pico micro SPI interface
- iii. Expected Output
 - Erase the Sector. Write/read the data's and o/p send to UART.

7. Gyroscopic and Accelerometer (MPU6050)

Write an application program to measure the Gyroscopic and Accelerometric values from MPU6050 module.

- i. Prerequisites
 - Pico Microcontroller
 - MPU6050
 - Bread board
- ii. Connection details
 - MPU6050 connected with I2C pins.
- iii. Expected Output
 - Read and write the configuration register values.
 - Based on configuration values, read the gyroscopic and accelerometric readings from the respective registers print in putty serial console.

8. Interrupt handler

Write a interrupt handler program to blink the LED whenever interrupt occurred.

- i. Prerequisites
 - Pico Microcontroller
 - Bread board
- ii. Connection details
 - Push button interface with Interrupt pin of micro
 - LED connected with GPIO pin
- iii. Expected Output
 - On press of push button, the LED should blink 5 times and then off.

9. CAN Communication.

Write a C program to communicate with the two CAN module (Master/Slave)

i. Prerequisites

Setup 1:

- Pico Microcontroller
- MCP2515
- Bread board

Setup 2:

- Pico Microcontroller
- MCP2515
- Bread board

ii. Connection details

Have a two set of CAN and Micro devices. Each CAN and Micro devices connected with SPI interface. CAN output signals are connected each others. Configure the one set is Transmitter and another one is Receiver.

iii. Expected Output

- Receiver receives the messages from transmitter and prints the messages.

10. Print messages in 16*2 LCD display

Write a program to print the messages in 16*2 LCD display.

i. Prerequisites

- Pico Microcontroller
- 16*2 LCD display

ii. Connection details

- LCD module interface with gpio's

iii. Expected Output

- Display your name in LCD.

11. Revision History

Rev. No.	Date (dd/mm/yy)	Section	Change Description	Modified by
0.1	21-09-2023	All	Initial contents	Durai (dpanneer)
0.2	12-05-2024	All	* Updated the 9th Experiment with CAN communication. * Updated the 6th Experiment with NOR interface.	Durai (dpanneer)